

MAEG4060 Virtual Reality Systems and Applications

Course Project

Design and develop an interactive 3D virtual environment (Not a video game, i.e. no mission, no user scores required). The application should be written using Visual C++, DirectX9.0c. Keyboard or gamepad control is expected. The virtual environment should include some physical simulation, e.g. objects falling according to gravity, vehicles accelerate and decelerate according to user control, moving on a terrain surface, interaction with objects in the virtual environment, etc.

You can choose to work on the project as an individual project or as a group project with at most two members.

You are to submit your work in 3 stages:

Report on 1st stage development

The report is to give a brief description or project proposal (a few pages) of the virtual environment to be developed. This includes

1. Description of the main theme of the virtual environment
2. Descriptions and sketches (hand sketches are accepted) of the models to be included in the virtual environment
3. Descriptions of the effects to be included (e.g. movement control, collision detection and responds, natural phenomenon, etc.)

Stage one report will serve as a reference for your project development. The actual project content can be revised in the development process. If the main theme of the project deviates much from the submitted stage one report at a later stage, a revised stage one report will have to be submitted again. Submit your work in a single PDF file.

Submission deadline: 23 Jan 2026

Report on 2nd stage development

In the second stage, all models of the virtual environment are to be created. Models should be tested for conversion to DirectX format. Program coding should also be started for testing whether the converted models can be imported into your program.

Materials to be submitted:

1. Height map of the landscape model if any, or 3D models of the textured landscape environment in Maya or DirectX file format
2. Textured 3D models (Maya or DirectX file) of all other objects (with the necessary animation for group project)

Submission deadline: 6 March 2026

Final report

Report content:

1. The completed program
2. A user guide
3. A brief description of the techniques used. If some of the 3D models used are downloaded from the web, then state the sources of the 3D models and any modifications made on the models.

Submission deadline: 17 April 2026

Marking Scheme:

Since the project content may be changed in the development process, the project work will be marked after the final report is submitted.

For group project, you are required to include model animation in your program. There can be at most 2 students in one group. Same marks will be given to students of the same group.

Stage	Task	Group Project	Individual Project
1	Proposal	10%	10%
2	3D Models of Landscape and Objects	20%	40%
	Animation (Created in Maya and is used in the VR application)	20%	
3	Gamepad or Joystick Control	5%	5%
	Model Navigation	5%	5%
	Collision Detection with static object	10%	10%
	Physics based object motion simulation	10%	10%
	User Guide	5%	5%
	Report	15%	15%

Bonus : (Animation(for individual project only), Collision Detection with moving objects, etc.) – 10%

Full marks (maximum): 100%